

Digest: Physics of the Riemann Hypothesis

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1 Introduction

This document digests the comprehensive review “Physics of the Riemann Hypothesis.” The paper bridges the Riemann zeta function $\zeta(s)$ with various physical phenomena, suggesting that the elusive proof of the Riemann Hypothesis might be encoded within the laws of physics.

2 Classical and Quantum Mechanics

The paper explores the Hilbert-Pólya conjecture, which posits that the non-trivial zeros of $\zeta(s)$ correspond to the energy eigenvalues of a self-adjoint Hamiltonian operator. Trace formulas, such as Gutzwiller’s and Selberg’s, link the distribution of prime numbers to the unstable periodic orbits of classically chaotic systems.

3 Random Matrix Theory and Nuclear Physics

The nearest-neighbor spacing distribution of the non-trivial zeros of the Riemann zeta function exhibits striking similarity to the eigenvalue spacing of random matrices drawn from the Gaussian Unitary Ensemble (GUE), a theory originally developed to model the energy levels of heavy atomic nuclei.

4 Statistical Mechanics: The Primon Gas

A fascinating mathematical model discussed is the “Riemann gas” or “primon gas.” In this model, non-interacting bosons (primons) occupy energy levels proportional to the logarithm of prime numbers, $E_p = E_0 \ln p$. The resulting canonical partition function is exactly the Riemann zeta function:

$$Z_B(T) = \sum_{n=1}^{\infty} e^{-E_n/k_B T} = \sum_{n=1}^{\infty} n^{-s} = \zeta(s) \quad (1)$$

where $s = E_0/k_B T$. This framework offers a direct physical manifestation of the zeta function's pole at $s = 1$, corresponding to the Hagedorn temperature limit.